

KONGU ENGINEERING COLLEGE

PERUNDURAI ERODE-638060



Transform Yourself

DEPARTMENT OF COMPUTER TECHNOLOGY-PG

M.Sc SOFTWARE SYSTEMS

A1 WATER TECH – DIGITAL SERVICE AND GST BILLING SYSTEM

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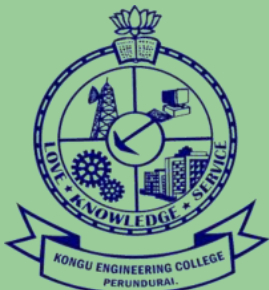
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Estd : 1984

Objective

The objective of this project is to digitize the operations of a service-based company by providing a customer website for service. Product ordering and an admin mobile application for **GST-based billing and invoice generation**. The system uses **React JS, Flutter, and Firebase** to ensure real-time data storage, editable records, and bilingual support (English and Tamil).

Estd : 1984

Existing System

- The existing system relies on manual billing, offline records, and in-person communication for managing services and products. There is no real-time data storage or automation.

Disadvantages:

- Time-consuming manual process
- GST calculation errors
- No centralized data management
- Limited scalability

Proposed System

- The proposed system is a digital service and billing platform designed to overcome the limitations of the traditional manual system by providing an integrated website and mobile application for service-based businesses.

Advantages:

- Online product and service ordering
- GST-based automated billing
- Editable records with real-time cloud storage
- PDF invoice generation
- Bilingual support (English & Tamil)

Modules

Admin Management

- Handles admin login, product/service control, billing, and system settings.

Customer Ordering

- Allows customers to view services/products and place orders through the website.

Billing & Invoice Management

- Generates editable GST bills, supports payment modes, and produces PDF invoices.

Service Recommendation Model (Apriori)

- Analyzes past orders, frequently selected services, and seasonal demand to recommend relevant services and products.

Admin Management

Manages products, services, orders, and system settings through secure admin access.

- Admin login and access control
- Product and service management
- Order monitoring
- System and language settings

Customer Ordering

Allows customers to view services and products and place orders through the website.

- Service and product listing
- Order placement
- Order details stored in Firebase
- Offline service fulfillment at shop

Billing & Invoice Management

Handles GST-based billing and invoice generation using the admin mobile application.

- Automatic GST calculation
- Payment mode selection (Cash / Net Banking)
- Editable bill records
- PDF invoice generation and storage

Service Recommendation Model (Apriori Algorithm)

Uses past order data to recommend frequently selected and seasonally popular services, and by using Apriori Algorithm.

- Analyzes past orders
- Identifies frequently combined services
- Supports seasonal demand prediction
- Helps admin suggest relevant services

Technologies Used

Front End :

- React JS
- Flutter
- NodeJS

Database :

- Firebase Firestore

Cloud Storage:

- Firebase Storage

Back End :

- Firebase Functions
- Firebase Authentication

Additional Features:

- PDF Generation
- GST Calculation Logic
- Bilingual Support (English & Tamil)